Interview Preparation - Top Questions

Answer length: answer_medium; Rounds: Technical round 1

1. Can you describe your experience architecting a high-scale rule-authoring platform at Amazon? What were the key challenges and how did you address them?

2. You mentioned engineering a specialized Java library. Can you elaborate on its purpose, the technical challenges involved, and how it achieved a 25-hour reduction in client latency?

3. How did you achieve an 85% reduction in server-side execution and $30,000 annual infrastructure cost savings through strategic operational caching?

4. You automated the deployment lifecycle to full CD, saving 260 developer-days annually. What tools and processes did you implement, and what were the main challenges?

5. Tell me about pioneering a local testing environment. What problem did it solve, and how did it accelerate development velocity and reduce costs?

6. You designed and delivered event-driven microservices using Java/Spring Boot on AWS. Can you explain the benefits of an event-driven architecture and how you implemented it?

7. How did you improve system scalability and performance by integrating AWS services like EC2, S3, Lambda, and DynamoDB?

8. You enhanced system observability by integrating distributed tracing and structured logging. What specific tools did you use, and how did this reduce MTTR?

9. Describe your experience spearheading DevOps best practices, including the 'test pyramid' and staging canaries, to ensure safe and frequent production deployments.

10. How do you approach debugging and monitoring applications using AWS CloudWatch Logs, and what specific techniques do you find most effective?

11. Explain the concept of idempotency in distributed systems and why it's crucial, especially when dealing with retries and event processing.

12. What is the 'saga pattern' in microservices, and when would you use it over a two-phase commit for managing distributed transactions?

13. Given your experience with Docker and Kubernetes, how would you containerize a Spring Boot microservice and deploy it to AWS EKS?

14. You have experience with Terraform. How does Terraform fit into your CI/CD pipeline for managing AWS infrastructure?

15. How do you ensure data consistency in a distributed system that uses DynamoDB, especially when dealing with eventual consistency?

16. Describe a challenging bug you encountered in a distributed system and how you debugged and resolved it.

17. How do you approach designing APIs for microservices, considering aspects like REST/gRPC, JSON/Protobuf, and versioning?

18. You mentioned using JUnit 5, Mockito, WireMock, and Testcontainers. How do these tools collectively contribute to a robust testing strategy for microservices?

19. What are the core principles of designing a resilient distributed system, and how have you applied them in your work at Amazon?

20. Explain the concept of circuit breaking in distributed systems and how you might implement it in a Spring Boot application?